Candidate Number ____________________ Score ____________________

The 29th International Biology Olympiad National Team Selection Camp in 2018

Second exam practice questions

I. Materials and Equipment:

project	quantity
Tablet PC	**1** unit
Engineering Computer	**1** unit
Different colored plant seeds (one tray each of A, B, and C)	**3** plates
white paper	**2** sheets

Precautions:

1. Please confirm that your candidate number is correct; if it is incorrect, please raise your hand and ask the teaching assistant for assistance.
2. Please check the materials and equipment on the table. If there are any errors, please raise your hand and ask the teaching assistant to handle them.
3. This examination room contains a question paper (4 pages), an answer sheet (4 pages), and two blank sheets of paper. Please return all of them when submitting your exam.
4. The time allotted for answering the questions is 90 minutes. Please answer on the answer sheet.
5. Answers may be written on the back of the questions, but please indicate the question number.

Part 1: Biomes (50 **points)**

In front of you are three petri dishes. Biological research requires good work habits. The number and composition of plant seeds in these three petri dishes are pre-set; please do not mix them up. At the end of the test, staff will count the number of plant seeds in each petri dish. If the count does not match the pre-set values, your score for this part will be deducted.
Score.

Each petri dish is considered a biological cluster, and each plant seed represents an individual organism. Seeds of similar color, shape, and size are considered to be of the same species, while those with significant differences in color, shape, and size are considered to be different organisms.
kind.

1. Please inventory all individuals (seeds) within cluster A (all plant seeds in petri dish A), classify them into different species, and give them simple names based on their characteristics (e.g., soybean). Please create a binary key (either a or b) on the answer sheet, using appropriate characteristics to distinguish all species in cluster A. (24 points)

2.1 Please fill in the number of individuals of each species within the three groups A, B, and C into Table 2.1 of the answer sheet. (3 points)

2.2 Please calculate the Mehinick richness index for clusters A, B, and C (accurate to 3 decimal places). Please fill in the answer sheet. (3 points)

Mehinick's richness index

$$= \frac{S}{ÿN}$$

R: Richness index

S: Total number of species

N: Total number of individuals

2.3 Which of the three clusters, A, B, and C, has the highest Shannon dissimilarity index? (1 point) What is the value of its Shannon dissimilarity index (accurate to 3 decimal places)? (2 points)

2.4 Which of the three clusters, A, B, and C, has the highest Pielou evenness index? (1 point) What is the value of its Pielou evenness index (accurate to 3 decimal places)? (2 points)

(Please fill in your answers on the answer sheet)

Shannon Dissimilarity Index

D = –ÿ(Piÿln (Pi))

D: Shannon Dissimilarity Index

Pi: The proportion of the number of individuals of the i-th type to the total number of individuals in the cluster, with a value between

0 and 1. ln: The natural logarithm.

Pielou's uniformity index E =

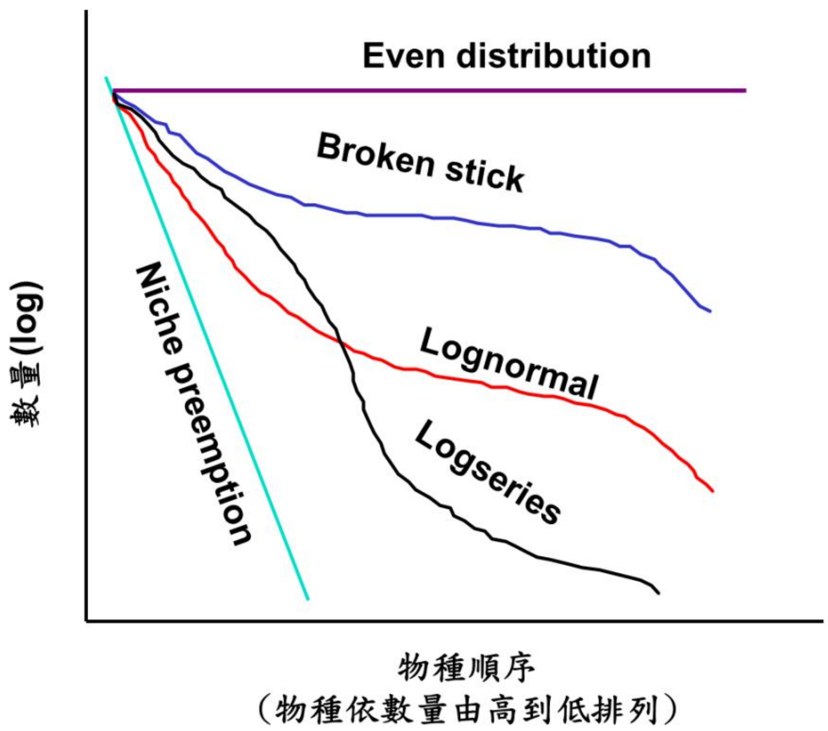
D / Dmax = D / ln(S)

E: Pielou's evenness index; D:

Shannon's dissimilarity index Dmax: The theoretical

maximum value of Shannon's dissimilarity index for this cluster, assuming the total number of species and individuals remains constant. This maximum value is the natural logarithm of the total number of species S, ln(S).

Besides the table above, another complete way to represent the population distribution of species within a cluster is through a rank-abundance diagram. A rank-abundance diagram sequentially lists the population proportions (Pi) of all species according to their dominance. The most dominant species are listed first, the second most dominant species second, and so on, down to the rarest species. The steeper the slope of the rank-abundance diagram, the greater the dominance of common species (a steeper slope indicates a larger difference in population proportions). The following figure shows a common pattern of species-population relationships in biological clusters (note that the vertical axis of this figure is logarithmic).



3.1 If the number of species in a cluster is determined to be S, and the total number of individuals is N, under which of the following patterns of species-number relationships can the lowest Shannon dissimilarity index be obtained? (A) Even distribution, (B) Broken stick, (C) Lognormal, (D) Logseries, (E) Niche preemption. (3 points)

3.2 Which species-quantity relationship pattern best fits cluster A? (A) Even distribution, (B) Broken stick,(C) Lognormal,(D) Log series,(E)Niche preemptionÿ(3 ÿ)

3.3 Which species-quantity relationship pattern best matches cluster C? (A) Even distribution, (B) Broken stick,(C) Lognormal,(D) Log series,(E)Niche preemptionÿ(3 ÿ)

4.1 In these three clusters, if legume seeds represent producers, cucurbit seeds represent primary consumers, and aster seeds represent secondary consumers, and the weight of individual seeds represents the biomass of each organism, based on the total biomass of these three trophic levels, which of the three clusters A, B, and C is most likely a grassland ecosystem? (1 point) Please explain your reasoning. (4 points)

Part Two: Photographic Observation of Primate Behavior (50 **points)**

In recent years, global conservation efforts for primates have led to an increase in the number of macaques worldwide, increasing their opportunities for interaction with humans. Many tourists are excited to observe the macaques, and some even feed them. However, this feeding behavior has sometimes caused conflicts between humans and monkeys. The following photos, taken by researchers during on-site observations, depict interactions between Formosan macaques and other monkeys from other countries, as well as the behavior of wild capuchin monkeys in the wild. Carefully observe the behaviors of the animals and humans in the pictures and explore their implications. Most of the photos in the series will be presented individually in larger sizes for easier observation and comparison.

1) Based on the two pictures A and B below, describe their meanings and compare the differences between the two pictures. (7 points)

2) Based on the picture below showing a human feeding a Formosan macaque, explain how you estimate the distance between the macaque and the human. What is the distance between the human and the macaque shown in this picture? Explain its meaning and discuss the possible effects of human feeding on the macaque. (10 points)

3) Based on the three pictures A, B, and C below, compare the differences in human behavior when feeding the monkey. (8 points) 4) Based on the pictures A and B below, both pictures show the behavior of the macaques when facing a photographer alone. Based on their appearance and behavior, explain which poses a greater threat to humans. (9 points) 5) Based on the pictures below, is there a problem of baboons coming into contact with humans in South African parks? Discuss the management methods of the authorities and their advantages and disadvantages. (8 points) 6) The five pictures (ABCDE) below show the behavior of capuchin monkeys captured by a photographer within one minute. Arrange the order in which they occurred. Explain how you determined the first behavior. What was the monkey doing? Describe the possible meaning of this behavior. (8 points)

(Please fill in your answers on the answer sheet)